

Inclusion Exclusion Criteria

Dimension	Inclusion Criteria	Exclusion Criteria
Publication Type	Peer-reviewed journal articles and conference papers	Non-peer-reviewed sources, including reports, theses, dissertations, editorials, book chapters, blogs, and opinion articles
Language	Publications written in English	Publications written in languages other than English
Publication Period	Studies published between 2019 and 2025	Studies published before 2019 or after 2025
Research Methodology	Empirical studies reporting datasets, predictive models, validation procedures, and evaluation results	Studies lacking sufficient methodological information, datasets, model descriptions, or evaluation metrics
Educational Context	Research conducted in educational settings, including LMS, MOOCs, online learning, blended learning, and classroom environments	Studies conducted outside educational or learning-related contexts
Research Focus	Studies investigating academic procrastination or closely related procrastination-oriented learning behaviours using machine learning, deep learning, educational data mining, learning analytics, or artificial intelligence techniques	Studies not addressing procrastination-related behaviours or not employing predictive analytics or machine learning methods